Smart Crop Recommendation System Using Decision Tree Classification and Full-Stack Web Deployment

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ABSTRACT

Food insecurity remains a critical global challenge addressed by SDG 2: Zero Hunger. This study develops a machine learning-based crop recommendation system trained on a multi-feature agricultural dataset. Three classifiers — Decision Tree, Random Forest, and Gaussian Naïve Bayes — were evaluated, all achieving 100% accuracy. The best model was deployed as a web application to provide farmers with data-driven crop recommendations based on soil, climate, and nutrient conditions.

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INTRODUCTION

Selecting the right crop for a given environment is essential for maximizing yield and ensuring food security. Traditional methods rely on experience and are prone to error. Machine learning offers a scalable, data-driven alternative. This project builds a crop recommendation system aligned with SDG 2, using soil chemistry, climate, and water data to recommend optimal crops through an accessible web interface.

METHODS AND MATERIALS

DATASET: 57,000 samples across 57 crop classes with 22 features including soil pH, temperature, humidity, NPK values, water requirements, and encoded categorical variables (soil type, season, water source).

PREPROCESSING: Label encoding of categorical features, class-balanced stratified train/test split (80/20).

MODELS TRAINED: Decision Tree, Random Forest (100 trees), and Gaussian Naïve Bayes.

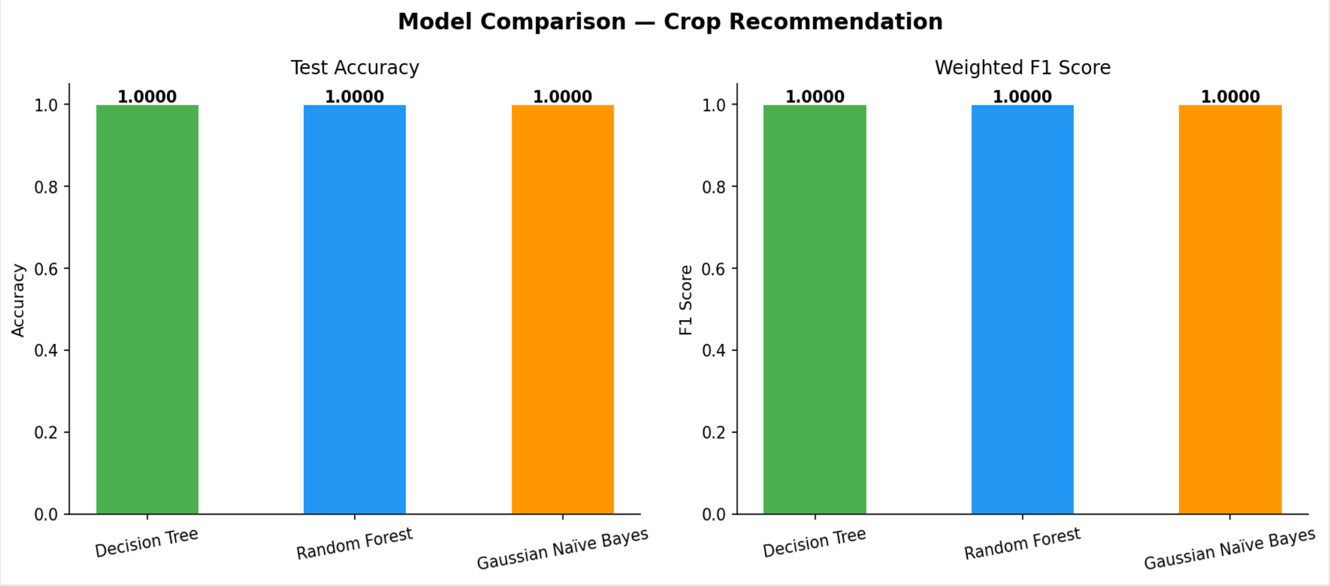
EVALUATION METRICS: Accuracy, Weighted F1-score, 5-fold cross-validation.

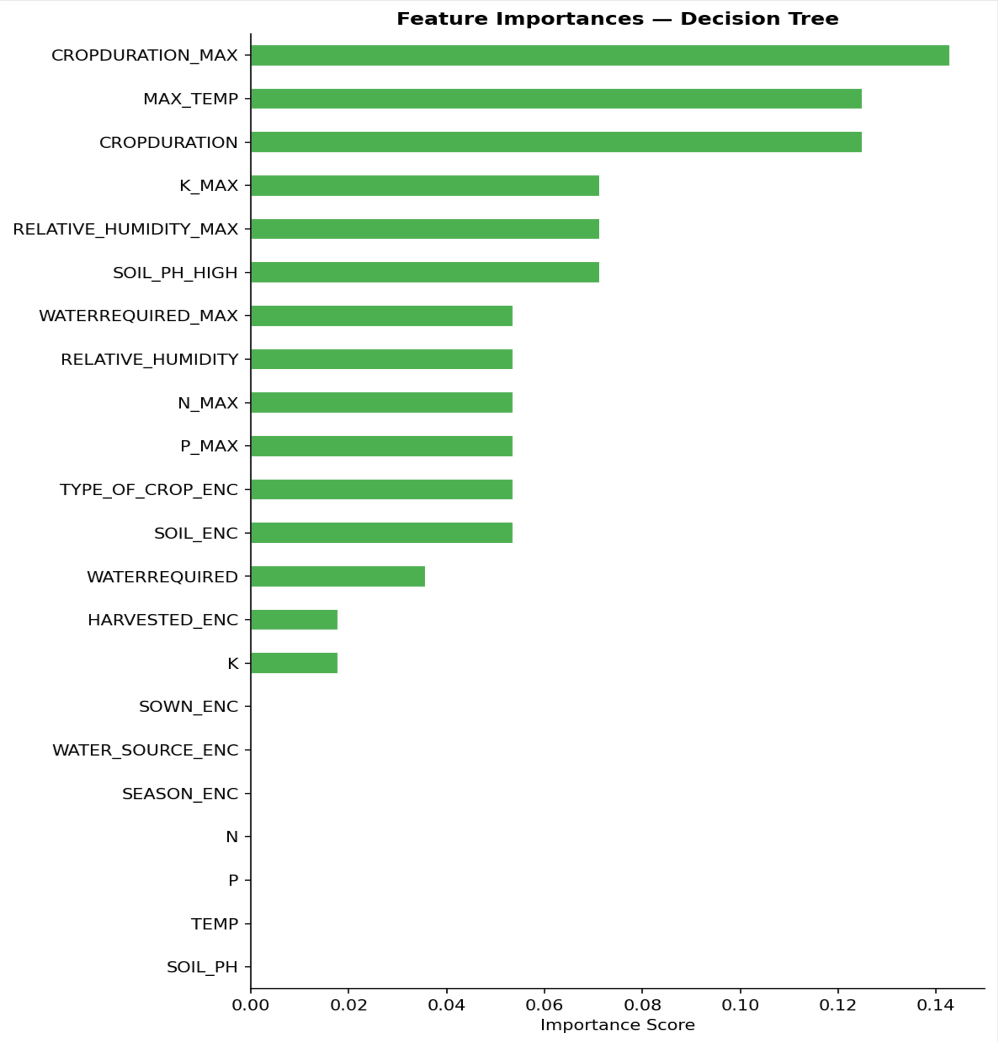
DEPLOYMENT: Best model serialized via Pickle and integrated into a Python web application.

RESULTS

All three models achieved 100% accuracy and weighted F1-score on the test set, indicating highly separable class boundaries in the dataset. Decision Tree was selected as the final model due to its interpretability and low inference overhead. 5-fold cross-validation confirmed consistent performance with no signs of overfitting.

Model	Accuracy	Weighted F1
Decision Tree	1.0000	1.0000
Random Forest	1.0000	1.0000
Gaussian Naïve Bayes	1.0000	1.0000





DISCUSSION

The perfect accuracy across all models suggests the dataset's features are highly discriminative for crop classification. The Decision Tree's interpretability makes it especially suitable for a public-facing tool, as its decisions can be traced back to specific thresholds (e.g. soil pH > 6.5). The web app lowers the barrier for non-technical users, directly supporting SDG 2 goals by making precision agriculture more accessible.

CONCLUSIONS

This project successfully demonstrates that machine learning can reliably recommend crops based on environmental and soil conditions. The deployed web application provides an accessible tool for data-driven agricultural decision-making. Future work could incorporate real-time sensor data, regional datasets, and a larger crop variety to improve generalizability.

REFERENCES

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